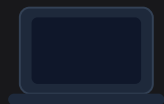


Setting Up Cloudflare Tunnel

Share your gallery with anyone, anywhere in the world — a complete, screenshot-by-screenshot walkthrough. No networking experience required.



Your Mac
localhost:8080

cloudflared



Cloudflare
encrypted tunnel

https://



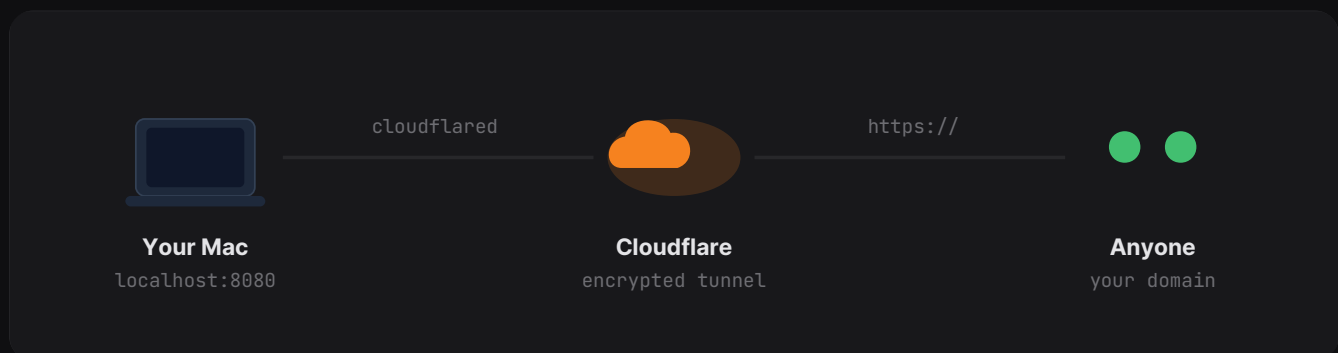
Anyone
your domain

BEFORE YOU BEGIN

What Cloudflare Tunnel does

By default, PIX only shares your gallery with people on your own Wi-Fi. Cloudflare Tunnel opens that up to anyone, anywhere — using a domain name you choose, secured with a real HTTPS certificate.

Normally, making something on your Mac reachable from the internet means configuring “port forwarding” on your router: a fiddly process, and one that leaves a door open on your network. A Cloudflare Tunnel avoids all of that. Instead of opening a door **in**, your Mac makes a single secure connection **out** to Cloudflare. Cloudflare then hands visitors off to that connection whenever they visit your domain. No router settings, no open ports, and it’s free.



Your Mac connects out to Cloudflare; Cloudflare connects visitors in. Nothing is ever opened on your router.

What you'll need

- PIX installed, with a folder selected and the server started at least once
- A free Cloudflare account — no credit card required
- A domain name you already own, added to that Cloudflare account
- About 10–15 minutes
- Comfort pasting a few commands into Terminal — we'll show you exactly what to type

Don't have a domain in Cloudflare yet?

Sign up free at dash.cloudflare.com/sign-up, then add your domain from the dashboard so Cloudflare manages its DNS. This can take a few minutes to become “Active” — once it shows Active, you're ready for Step 1.

New to Terminal? Start here.

A couple of steps in this guide ask you to run a command in Terminal. If you've never used it, here's everything you need — you won't have to memorize anything.

- 1 **Open Terminal.** Press `⌘ Space` to open Spotlight, type `Terminal`, then press `Return`.
- 2 **Copy the command** from this guide (select the text, or use the copy button in PIX's own Setup Guide window).
- 3 **Paste it in** with `⌘ V`, then press `Return` to run it.
- 4 **Read the output.** Each step below shows you exactly what a successful result looks like, so you can tell right away if something's off.

About the example screenshots

The Terminal screenshots in this guide show a prompt like `user@MacBook-Pro ~ %`. That part is just an example — yours will show your Mac's username and computer name instead. Likewise, paths like `/Users/user/` are placeholders for your own home folder. You never need to type the prompt itself; only the command that follows it.

You can't break anything

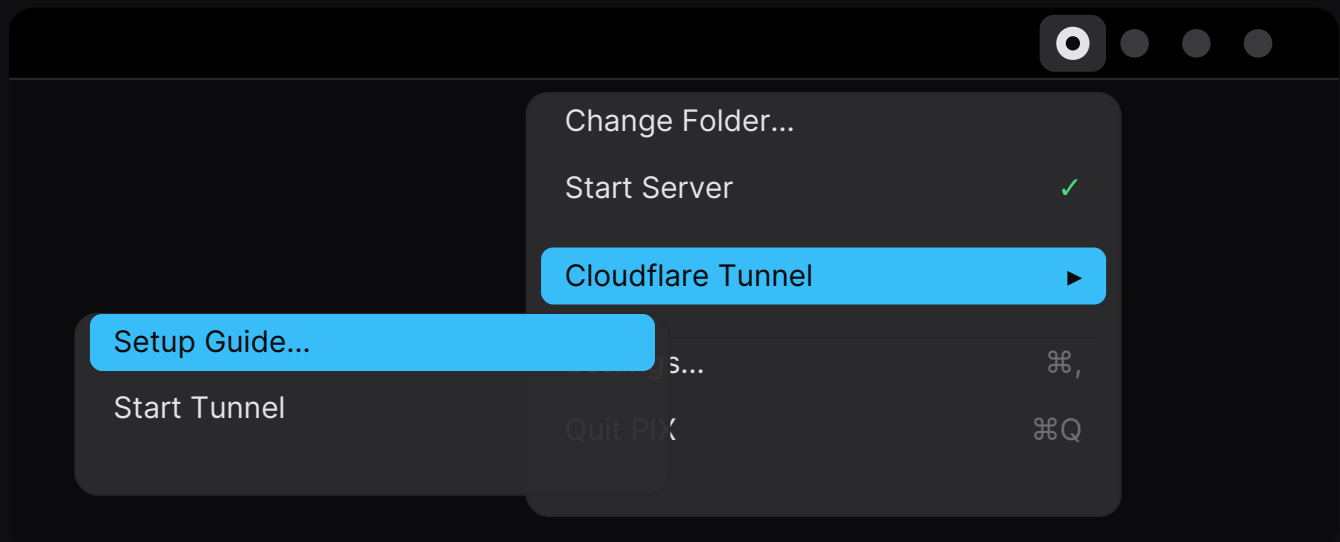
Every command in this guide only talks to Cloudflare or reads a small config file — none of them touch your photos, your system, or anyone else's files. If a command doesn't work, the worst case is simply trying it again.

1 Install cloudflared

cloudflared is Cloudflare's small helper program that creates the tunnel. PIX includes a one-click download for it — you don't need Homebrew or any developer tools.

Open the menu bar icon and choose **Cloudflare Tunnel → Setup Guide...**, then click **Download cloudflared**. Once it downloads, double-click the **.pkg** file and follow the installer prompts.

PIX MENU BAR

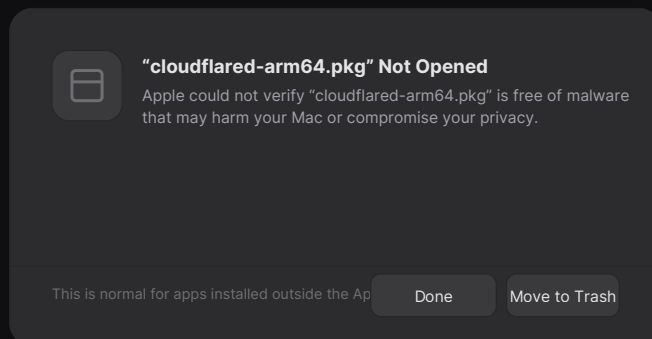


Menu bar icon → Cloudflare Tunnel → Setup Guide...

macOS will block the installer — this is expected

Because cloudflared is downloaded outside the Mac App Store, macOS shows a warning the first time. This isn't a virus alert or anything to worry about — it's just Apple being cautious about new downloads. Here's how to get past it.

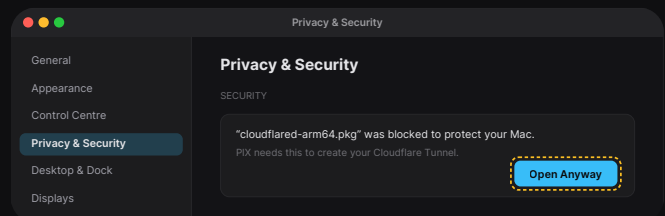
FIRST LAUNCH



1. You'll see this the first time you open the installer. Click Done.

Re-open (double-click) the **.pkg** installer once more — it will run normally this time.

SYSTEM SETTINGS



2. System Settings → Privacy & Security → Open Anyway. Enter your password if asked.

2 Log in to Cloudflare

This tells cloudflared which Cloudflare account — and which domain — it's allowed to create tunnels for.

In Terminal, run:

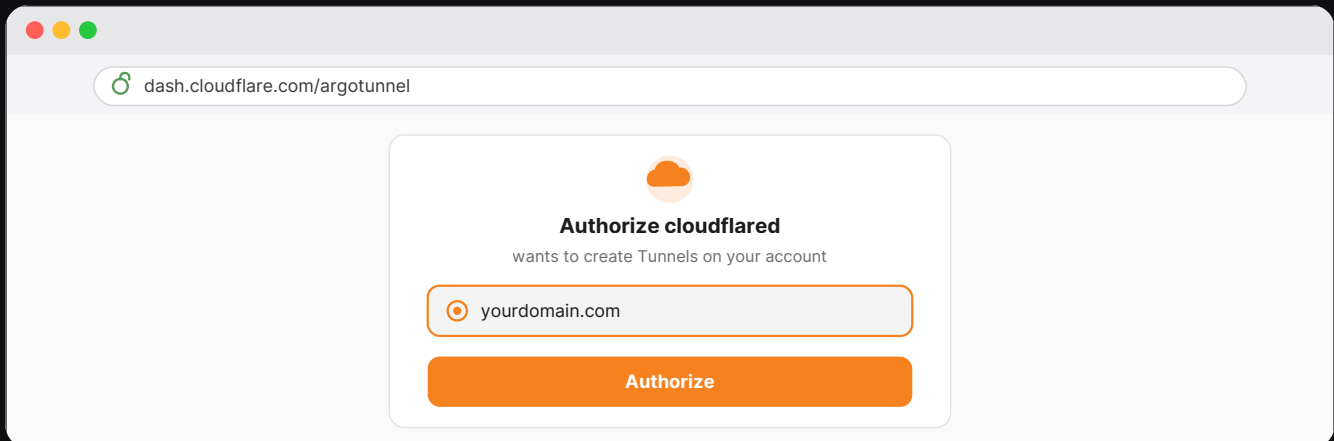
```
Terminal — zsh — 80x24

user@MacBook-Pro ~ % cloudflared tunnel login
Please open the following URL and log in with your Cloudflare account:
https://dash.cloudflare.com/argotunnel?callback=...
Leave cloudflared running to download the cert automatically.
```

cloudflared prints the login URL, then waits. Leave this window open — it confirms when you're logged in. (If it says "Failed to fetch resource" instead, see Troubleshooting.)

Your browser should open that page automatically. Choose the domain you want for your gallery, then click **Authorize**.

BROWSER



Have more than one domain? Pick whichever you want people to visit — you can always set this up again for another one later.

If your browser doesn't open — open the link yourself

Don't re-run the command. Leave Terminal running, select and copy the `https://dash.cloudflare.com/argotunnel?callback=...` URL it printed, and paste it into a browser window yourself — Terminal picks up the result on its own. Two things matter about that window:

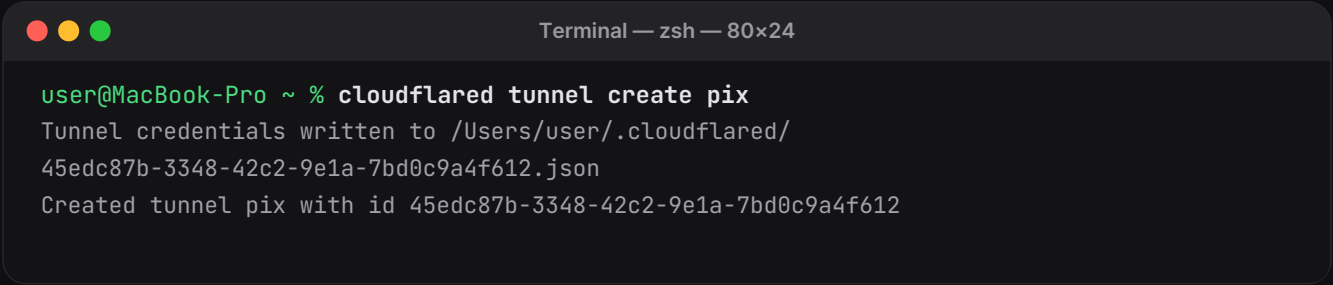
- **No ad blocker or privacy extension** — these frequently break the login handoff. A fresh Safari window, or a private window with extensions disabled, is safest.
- **No VPN or DNS filtering** (Pi-hole, NextDNS, AdGuard, a corporate network). Pause it for the minute this takes, then turn it back on afterwards.

3

Create a tunnel

Now you'll create the actual tunnel — think of it as registering a private, named channel between your Mac and Cloudflare.

In Terminal, run:

A screenshot of a macOS Terminal window. The title bar at the top says "Terminal — zsh — 80x24". The window has three colored window control buttons (red, yellow, green) on the left. The terminal text shows a user at a MacBook-Pro prompt running the command "cloudflared tunnel create pix". The output shows that credentials were written to a file in the user's home directory and that a tunnel named "pix" was created with a specific ID.

```
user@MacBook-Pro ~ % cloudflared tunnel create pix
Tunnel credentials written to /Users/user/.cloudflared/
45edc87b-3348-42c2-9e1a-7bd0c9a4f612.json
Created tunnel pix with id 45edc87b-3348-42c2-9e1a-7bd0c9a4f612
```

The long code after “with id” is your Tunnel ID — you’ll need it in the next step.

Copy your Tunnel ID somewhere safe

It’s the long string that looks like `45edc87b-3348-42c2-...` — select it and copy it into Notes or TextEdit for a moment. You’ll paste it twice in the next step.

This also quietly saves a small credentials file to `~/.cloudflared/` on your Mac — that’s what proves to Cloudflare that this tunnel belongs to you. You won’t need to touch it directly.

4 Edit the config file

The config file is what tells cloudflared “when someone visits this address, send them to PIX running on this Mac.”

Back in PIX’s Setup Guide window, click **Open Config File...** This opens (or creates) `~/.cloudflared/config.yml` in your default text editor. Replace the highlighted values:

CONFIG.YML

```

1  # fill in your Tunnel ID (twice) and your hostname
2
3  tunnel: YOUR-TUNNEL-ID
4  credentials-file: /Users/yourname/.cloudflared/YOUR-TUNNEL-ID.json
5
6  ingress:
7    - hostname: pix.yourdomain.com
8    service: http://localhost:8080
9    - service: http_status:404

```

Amber values: your Tunnel ID (appears twice). Blue values: your Mac username and hostname.

tunnel:	Your Tunnel ID from Step 3.
credentials-file:	Same Tunnel ID again, plus your macOS username in place of <code>yourname</code> . The <code>.json</code> filename must match exactly what Step 3 printed.
hostname:	Whatever address you want people to visit — e.g. <code>pix.yourdomain.com</code> or <code>photos.yourdomain.com</code> . Keep this the same in Step 5.
localhost:8080	The port PIX is running on. Leave this as 8080 unless you’ve changed PIX’s port in Settings.

Both Tunnel ID fields must match

The ID next to `tunnel:` and the one inside `credentials-file:` need to be identical — copy-paste the same value into both rather than retyping it.

5 Point your domain to the tunnel

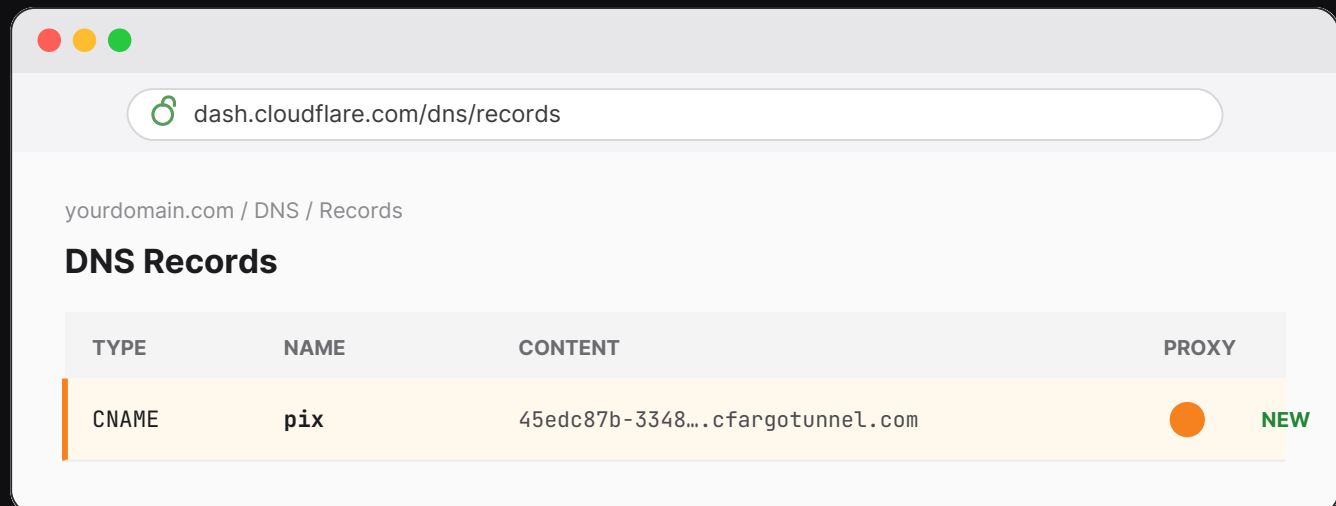
One last piece: telling Cloudflare's DNS that your chosen address should actually point at this tunnel.

Using the **same hostname** you put in the config file, run (replacing the example domain with yours):

```
Terminal — zsh — 80x24  
user@MacBook-Pro ~ % cloudflared tunnel route dns pix pix.yourdomain.com  
Added CNAME pix.yourdomain.com which will route to this tunnel
```

This creates the DNS record automatically — no manual editing needed.

CLOUDFLARE DASHBOARD



The screenshot shows the Cloudflare dashboard at `dash.cloudflare.com/dns/records`. The breadcrumb is `yourdomain.com / DNS / Records`. The title is **DNS Records**. Below is a table with the following data:

TYPE	NAME	CONTENT	PROXY
CNAME	pix	45edc87b-3348...cfargotunnel.com	 NEW

The new CNAME record appears in your Cloudflare DNS dashboard within moments.

If the command says a record already exists

Open the Cloudflare dashboard, go to **DNS → Records**, and add a CNAME by hand: **Name** is your subdomain (e.g. `pix`), **Target** is `<your-tunnel-id>.cfargotunnel.com`.

6

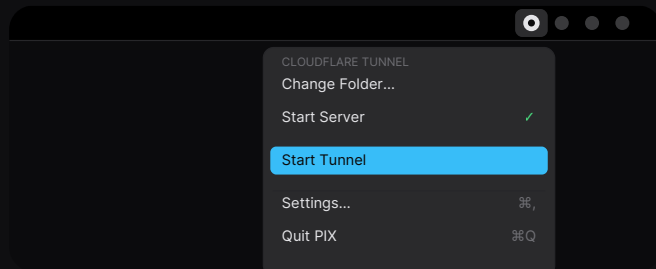
SIX STEPS TO A LIVE GALLERY

Start the tunnel

This is the part you'll actually use day to day — and it's a single click.

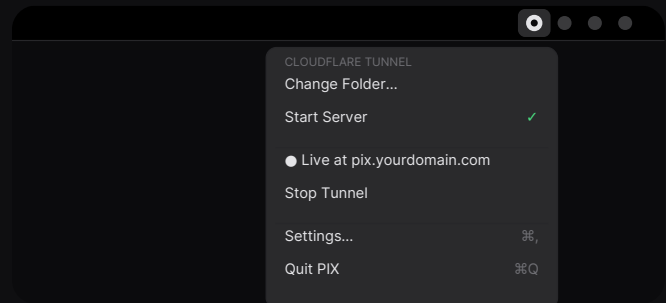
Make sure the PIX server is running, then open the menu bar icon and click **Start Tunnel**.

PIX MENU BAR



Click Start Tunnel from the Cloudflare Tunnel submenu.

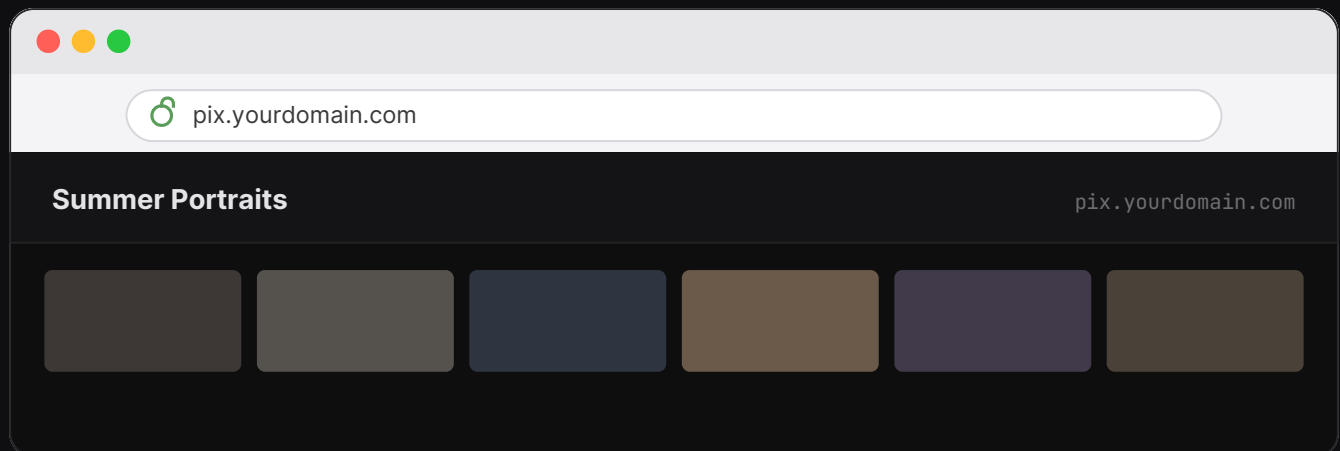
PIX MENU BAR, LIVE



A green dot means the tunnel is live and reachable from anywhere.

Within a few seconds, your gallery goes live at your domain:

YOUR LIVE GALLERY



Anyone with the link can now view your gallery — from anywhere.

That's it — you're live

Click **Stop Tunnel** any time to take your gallery private again; it'll still work fine on your local Wi-Fi. Starting and stopping never affects your DNS or config, so it's safe to toggle whenever you like.

Troubleshooting

These are the exact messages PIX shows when something's wrong, matched to what causes them and how to fix each one.

"cloudflared Not Found"

Cause: PIX looked in the two usual install locations and couldn't find cloudflared — usually because Step 1's installer hasn't fully completed yet.

Fix: Go to **System Settings → Privacy & Security** and click **Open Anyway** next to the cloudflared message (see Step 1), then quit and reopen PIX.

"cloudflared Blocked by macOS"

Cause: cloudflared is installed, but macOS's Gatekeeper is still preventing it from running — a quarantine flag that's sometimes left over even after clicking Open Anyway once.

Fix: Open Terminal and run the command PIX shows you in the alert:



Terminal — zsh — 80×24

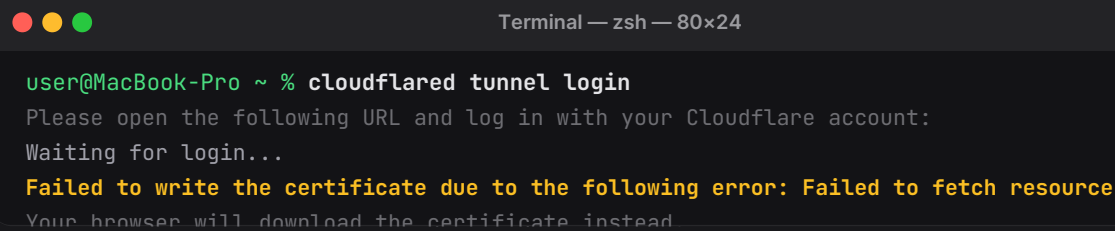
```
user@MacBook-Pro ~ % xattr -cr /usr/local/bin/cloudflared
```

Clears the quarantine flag directly. Then quit and reopen PIX.

If that doesn't help, go back to **System Settings → Privacy & Security** and click **Open Anyway** again.

Stuck after `cloudflared tunnel login` with “Failed to fetch resource”

Cause: After you authorize in the browser, it tries to hand a small certificate file back to cloudflared automatically. Something between the browser and cloudflared — usually an ad blocker, a privacy extension, a VPN, or firewall software — is blocking that local handoff.

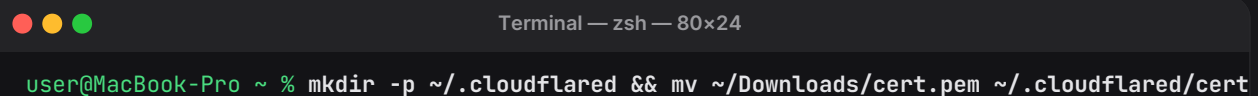


```
Terminal — zsh — 80x24

user@MacBook-Pro ~ % cloudflared tunnel login
Please open the following URL and log in with your Cloudflare account:
Waiting for login...
Failed to write the certificate due to the following error: Failed to fetch resource
Your browser will download the certificate instead.
```

When the automatic handoff fails, cloudflared says so and falls back to a direct download.

Fix: Check your **Downloads** folder for a file named `cert.pem` — that’s the fallback download mentioned above. If it’s there, move it into place:



```
Terminal — zsh — 80x24

user@MacBook-Pro ~ % mkdir -p ~/.cloudflared && mv ~/Downloads/cert.pem ~/.cloudflared/cert
```

Then run `cloudflared tunnel login` again — it should complete instantly this time.

If no file downloaded either, temporarily turn off ad blockers or privacy extensions (or try Safari instead), and if you’re on a VPN or run firewall software like Little Snitch, pause it briefly — then run the login command again.

“Tunnel Failed to Start” or the tunnel won’t start the first time

Cause: Almost always the same Gatekeeper quarantine issue above, right after a fresh install.

Fix: Try the two fixes above, then **restart your Mac**. macOS often needs a reboot after installing a new command-line tool before it will run it — a restart also clears up any DNS that hasn’t taken effect yet.

“Tunnel Stopped” right after starting

Cause: cloudflared started, then quit on its own — almost always a small mismatch in the config file.

Fix: Reopen `config.yml` (Step 4) and check that:

- The Tunnel ID appears identically in both the `tunnel:` and `credentials-file:` lines
- The `.json` filename matches exactly what Step 3 printed
- Your macOS username is spelled correctly in the credentials path

Your domain doesn't load, or shows an error

Cause: DNS changes can take a few minutes to spread across the internet, especially right after Step 5.

Fix: Wait a few minutes and try again. Double check the domain shows **Active** in your Cloudflare dashboard, and that the CNAME record from Step 5 is there under **DNS → Records**.

It worked, then later stopped working after changing PIX's port

Cause: the config file still points at the old port.

Fix: Open `config.yml` and update `localhost:8080` to match PIX's current port, shown in **Settings → Server**.

FOR LATER

Quick reference

Every command from this guide, in one place.

COMMAND	WHAT IT DOES
<code>cloudflared tunnel login</code>	Connects cloudflared to your Cloudflare account (Step 2)
<code>cloudflared tunnel create pix</code>	Creates the tunnel and prints your Tunnel ID (Step 3)
<code>cloudflared tunnel route dns pix <hostname></code>	Points your domain at the tunnel (Step 5)
<code>cloudflared tunnel list</code>	Shows tunnels already created on this account
<code>xattr -cr /usr/local/bin/cloudflared</code>	Clears a macOS quarantine flag blocking cloudflared
<code>mv ~/Downloads/cert.pem ~/.cloudflared/cert.pem</code>	Manually places the login certificate if the automatic handoff fails

Where things live

FILE	LOCATION
Tunnel config	<code>~/.cloudflared/config.yml</code>
Tunnel credentials	<code>~/.cloudflared/<tunnel-id>.json</code>
cloudflared binary	<code>/usr/local/bin/cloudflared</code>

Checking your tunnel is still healthy

A green dot next to **Live at ...** in the PIX menu bar means the tunnel is up. If you ever want to see all tunnels tied to your account, run `cloudflared tunnel list` in Terminal.

Terms used in this guide

Terminal	A Mac app for typing commands directly to your computer, instead of clicking buttons. Find it via Spotlight (⌘ Space).
Tunnel	A private, secure connection your Mac opens out to Cloudflare, which Cloudflare then uses to forward visitors to your gallery.
Tunnel ID	A unique code Cloudflare generates for your tunnel when you create it in Step 3.
Config file	A small text file (<code>config.yml</code>) that tells cloudflared which address should connect to which app on your Mac.
Credentials file	A file Cloudflare creates alongside your tunnel that proves it belongs to your account.
DNS	The internet's address book — it's what translates a domain name like <code>yourdomain.com</code> into the right destination.
CNAME record	A single entry in your domain's DNS that says "this subdomain points to that destination." Step 5 creates one automatically.
Hostname	The address people will actually type or click to reach your gallery, e.g. <code>pix.yourdomain.com</code> .
Port	A number PIX uses to identify its own local connection on your Mac (8080 by default). Not something you normally need to think about, unless you've changed it in Settings.
Gatekeeper	macOS's built-in safety check for newly downloaded apps. It's why you see a warning the first time you open cloudflared's installer.

STILL STUCK?

Getting more help

If you've worked through the Troubleshooting section and your gallery still isn't reachable, here's where to go next.

fainimade.com

General questions, licensing, and support for PIX.

Cloudflare's own docs

For anything specific to your Cloudflare account or DNS settings, Cloudflare's support docs at developers.cloudflare.com go deeper than this guide covers.